Building a QuantLib Derivatives Workbench

Human-directed development with Codex and Claude, from swaps to callable XCCY

Tim Launer + two coding agents

Evidence through 22 August 2026

The product and its collaborative method matured together

Mar 2025 rates demo → Sep 2025 API repair → 3 Apr 2026 SOFR rebuild → 5–7 Apr models + risk → 7–14 Apr web platform → 14–21 Aug callable XCCY

- Git records the major releases; local birth times refine the one-commit August implementation sequence.
- Tim directed the economics; Codex built most specifications, code, integrations, and tests; Claude added review, validation, and regression depth.

Takeaway. The repository evolved with the human–agent team: every product increment also strengthened the method used to build the next one.

A pricer is a disciplined cash-flow machine

- The product defines dated, signed cash flows and exercise rights; the market supplies curves, volatility, FX, correlations, and fixings.
- The model maps uncertainty into state dynamics; the numerical method converts those dynamics into values and risks.
- Dates, currencies, day counts, collateral, quote conventions, and measures are economic inputs rather than metadata.

Takeaway. Sophisticated numerics cannot repair an incorrectly specified contract.

The first demo became a corrected QuantLib composition

- The March 2025 release connected a simple curve, swap, European swaption, and Bermudan through Python bindings.
- The initial Bermudan incorrectly used a European Black engine and an unsuitable underlying swap definition.
- September repairs introduced a forward-starting underlying and Hull–White `TreeSwaptionEngine`, while fixing changed QuantLib APIs.

Takeaway. Compact examples become reliable only when product, engine, conventions, and smoke tests agree.

SOFR curve construction became a tested calibration

Market construction

Overnight front point plus SOFR OIS helpers spanning short tenors to 30 years; log-cubic discount interpolation and explicit settlement conventions.

Evidence

Every helper reprices to its quote; discount, zero, and one-day forward views expose curve shape and interpolation behavior.

Takeaway. The curve is a calibrated market object with residuals, not merely a line drawn through rates.

European swaptions calibrate the short-rate model

- A diagonal strip maps each future exercise date to the tenor of its remaining swap.
- Normal-volatility quotes become QuantLib swaption helpers priced with a Jamshidian engine under Hull–White.
- Mean reversion is governed; a constant volatility minimizes declared helper pricing errors and reports every residual.

Takeaway. Calibration is reproducible only when helper selection, units, objective, bounds, and stopping rules are visible.

Bermudan pricing uses backward induction

$$V(t_i, x) = \max\left(E(t_i, x),\ \mathbb{E}[D_{i,i+1} V(t_{i+1}, X_{i+1}) \mid X_i = x]\right).$$

- Hull–White contributes a one-dimensional Gaussian short-rate state and an exact fit to today's curve.
- A recombining lattice propagates continuation values backward across business-day-adjusted exercise dates.
- Tree steps are a numerical parameter and require convergence checks.

Takeaway. The tree is natural for a single-currency, one-factor stopping problem, not a universal callable engine.

Model comparison, Greeks, and cliquets broadened challenge

- G2++ added a second Gaussian rates model; governed curve, volatility, and parameter bumps exposed calibration dependency.
- An SPX cliquet added Black–Scholes forward-start values, analytic Greeks, Monte Carlo distributions, and path-dependent scheduling.
- Parity, decomposition, scenario, common-random-number, and bump tests checked signs, scaling, and nonlinear response.

Takeaway. New products mattered because they expanded the repository's validation vocabulary, not merely its menu.

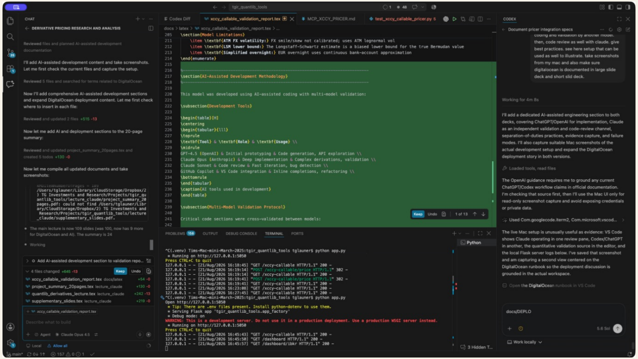
A small technical stack became the agents’ shared workbench

Flask routes and templates → Normalized JSON market + deals → QuantLib + Python pricing core → Tests, reports, CLI + MCP

- On the Mac, Git makes Codex and Claude changes, specifications, diffs, tests, JSON evidence, and server logs mutually inspectable.
- QuantLib supplies proven C++ primitives; Python makes the extension transparent; Apache, Gunicorn, and systemd run it on DigitalOcean.

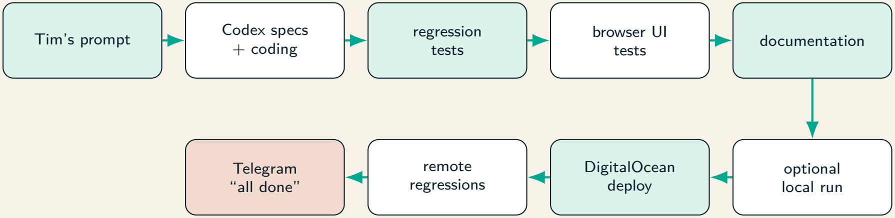
Takeaway. One repository connects human direction, two agents, open-source quant infrastructure, local evidence, and hosted execution.

Tim and two coding agents share one governed Mac workspace



Takeaway. This is where Tim guides, Codex builds, Claude reviews and extends validation, and all three inspect the same source and evidence.

One prompt can run the complete delivery chain



Takeaway. No manual handoff is required after the prompt; Claude's validation extensions and remote regressions protect later Codex changes.

DigitalOcean keeps production deliberately small

HTTPS client → Apache + TLS → Gunicorn 127.0.0.1:8008 → Flask + QuantLib

- systemd runs the app under `www-data` at `/var/www/html/tgir_quantlib_tools`; secrets stay in a mode-600 `.env`.
- Deployment builds a fresh final-path virtualenv, restarts the service, checks loopback and public TLS health, and rolls back on failure.

Takeaway. For about USD 7/month, this serious hobby gains a hosted evidence lane with TLS, process, health-check, regression, and rollback controls.

QuantLib lacks a native callable XCCY engine

Native building blocks

Curves, calendars, schedules, XCCY swaps, Hull–White/G2, swaption helpers and engines, stochastic processes, and Gaussian path generators.

Missing composition

A domestic-measure joint rates/FX process, callable cross-currency cash-flow semantics, and a multi-date optimal-stopping engine.

Takeaway. Native support is a spectrum: reuse mature primitives while owning the missing economics.

The benchmark deal fixes the economic question

- The example is a ten-year EUR/USD constant-notional swap, non-callable for two years and then annually cancellable.
- The holder receives quarterly compounded EUR ESTR and pays semiannual USD fixed at 4%.
- USD 100 million and EUR 90.909 million notionals exchange initially and finally; USD is collateral and reporting currency.

Takeaway. Exercise means cancelling the remaining tail under explicit event ordering, not receiving a generic option payoff.

The hybrid model couples two rates factors with FX

$$\begin{aligned} dr_d &= [\theta_d(t) - a_d r_d]dt + \sigma_d\, dW_d, \\ dr_f &= [\theta_f(t) - a_f r_f - \rho_{fS}\sigma_f\sigma_S(t)]dt + \sigma_f\, dW_f, \\ d\log S &= [r_d - r_f - \tfrac{1}{2}\sigma_S^2(t)]dt + \sigma_S(t)\, dW_S. \end{aligned}$$

- Dynamics are under the USD money-market measure with $S =$ USD per EUR.
- The full 3×3 Brownian correlation matrix must be positive semidefinite.

Takeaway. FX quotation and numeraire determine both the FX drift and the sign of the foreign quanto correction.

Calibration separates curves, rates vol, and FX vol

- USD SOFR discounts domestic cash flows; EUR ESTR forecasts coupons; FX forwards imply the USD-collateralized effective EUR discount curve.
- One constant Hull–White sigma per currency fits diagonal normal swaptions with governed mean reversions.
- Piecewise FX sigmas reproduce selected ATM Black maturities after including stochastic-rate contributions to terminal log-FX variance.

Takeaway. The exact relative RMS price error and instrument-level residuals disclose the cost of the fixed-volatility approximation.

Exact Gaussian transitions control Monte Carlo bias

- The state contains both Hull–White factors, log FX, and the integrated domestic short rate.
- Each interval uses exact Ornstein–Uhlenbeck decay and an integrated covariance matrix; eigenfactorization handles near-singular cases.
- The event grid contains coupon, fixing, payment, notional, exercise, and volatility-bucket dates, with intermediate step caps.

Takeaway. Exact conditional transitions reduce discretization bias but do not eliminate sampling error or model risk.

Two-pass LSM separates policy learning from pricing

1. Simulate training paths and move backward through the annual call dates.
2. Regress discounted future cancellable-tail value on standardized rates, log FX, leg values, squares, and cross-products.
3. For zero-fee cancellation, exercise when estimated continuation value is negative.
4. Freeze the policy and apply it forward on an independently seeded pricing sample.

Takeaway. Out-of-sample policy valuation removes in-sample look-ahead optimism and reports an honest lower-bound estimate.

The pricer and its validation framework can evolve together

Research evidence

Stored callable NPV is near USD 8.34 million versus non-callable NPV near USD 0.69 million. Residuals, martingales, exercise profiles, convergence, and bump stability qualify the mark.

Collaborative engineering path

Codex extends the specified core; Claude's regression layer challenges changes; Python keeps evidence transparent, while profiled C++ can add native QuantLib types and scalable simulation.

Takeaway. The system is designed to extend QuantLib rigorously: stabilize economics and tests first, then migrate measured bottlenecks to C++.

The compounding achievement

Tim directs $\rightarrow$ Codex specifies and builds $\rightarrow$ Claude reviews and extends validation $\rightarrow$ Mac proves $\rightarrow$ DigitalOcean hosts

A serious hobby: rigorous quantitative engineering from one prompt to tested deployment and a Telegram "all done"—for about USD 7/month.